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United States Patent Application Publication

20250282582

Kind Code

A1

Publication Date

September 11, 2025

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APPARATUS AND METHOD FOR REBUILDING A SPREADER BEAM

Abstract

A spreader beam and related method provide a simplified repair or rebuilding capability for wear parts of the spreader beam. The disclosed spreader beam includes removable sheave nests containing sheaves, which are the wear parts in a spreader beam. The disclosed spreader beam saves time and costs since an entire spreader beam need not be transported to a repair shop, rather only the removable sheave nests, or only the sheaves. In one embodiment, sheaves and sheave nests may be removed and replaced without the need for unwinding supporting cables.

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Appl. No.: 18/891953

Filed: September 20, 2024

Related U.S. Application Data

us-provisional-application US 63561804 20240306

Publication Classification

Int. Cl.: B66C1/66 (20060101); B66C1/12 (20060101)

U.S. Cl.:

CPC B66C1/663 (20130101); B66C1/125 (20130101);

Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. provisional application No. 63/561,804 filed 6 Mar. 2024, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure is generally directed to a spreader beam. More particularly, this disclosure is directed to a spreader beam with separable sheaves. Even more particularly, this disclosure is directed to a method of rebuilding, repairing or reconditioning a spreader beam.

BACKGROUND ART

[0003] Spreader beams are used in various applications, including foundries where molten iron or other metals are poured, for example, into sand molds. Other types of loads may also be borne by a spreader beam. A ladle or other load is held by the spreader beam, which in turn is lifted by a crane. The spreader beam includes at least two pullies also known as sheaves, or pluralities of sheaves, at opposite ends of an elongate beam. Cables or wire rope connect the sheaves to the crane.

[0004] The sheaves are wear parts that must be periodically repaired or replaced. Known spreader beams include sheaves that are integrally fastened to the elongate beam. In order to rebuild or repair the sheaves, the entire spreader beam, including the sheaves, must be transported to a repair shop. Cables must be unwound and removed. A typical spreader beam may weigh in excess of ten tons, thereby incurring significant transport cost.

SUMMARY OF THE INVENTION

[0005] The disclosed spreader beam provides a simplified repair or rebuilding capability for wear parts thereof. In particular, the spreader beam of the invention, because it includes removable sheave nests, can be rebuilt or repaired at lower cost. This is because only the sheave nests, and not the entire spreader beam (having integral sheaves) need to be shipped to a repair shop.

[0006] In one aspect, an exemplary embodiment of the present disclosure may provide a spreader beam comprising: a main frame adapted to suspend a load of material; and at least one sheave nest removably engaged with the main frame. In this exemplary embodiment or another exemplary embodiment, the spreader beam further comprises: at least one receiver operatively engaged with the main frame and configured to receive the at least one sheave nest. In this exemplary embodiment or another exemplary embodiment, the at least one receiver comprises: at least one side wall operatively engaged with the main frame; and at least one cradle plate operatively engaged with the at least one side wall, wherein the at least one cradle plate is configured to receive the at least one sheave nest. In this exemplary embodiment or another exemplary embodiment, the at least one receiver further comprises: a lower receiver aperture defined in the at least one side wall. In this exemplary embodiment or another exemplary embodiment, the at least one sheave nest comprises: at least one nest flange; at least one nest plate; a lower sheave nest aperture defined in at least one of the at least one nest flange and the at least one nest plate; and at least one sheave rotatably engaged with at least one of the at least one nest flange and the at least one nest plate. In this exemplary embodiment or another exemplary embodiment, the lower sheave nest aperture of the at least one sheave nest is coaxial with said lower receiver aperture of the at least one receiver when the at least one sheave nest is engaged with the at least one receiver. In this exemplary embodiment or another exemplary embodiment, the spreader beam further comprises: at least one

lower securement assembly operatively engageable with the at least one sheave nest at the at least one lower sheave nest aperture and with the at least one receiver at the lower receiver aperture. In this exemplary embodiment or another exemplary embodiment, the at least one sheave nest further comprises: at least one nest plate operatively engaged with the at least one nest flange; at least one upper sheave nest aperture defined by at least one of the at least one nest flange and the at least one nest plate. In this exemplary embodiment or another exemplary embodiment, the at least one receiver further comprises: at least another side wall operatively engaged with the main frame and opposite to the at least one side wall; and an upper receiver aperture defined in at least one of the at least one side wall and the at least another side wall. In this exemplary embodiment or another exemplary embodiment, the upper receiver aperture is coaxial with the at least one upper sheave nest aperture when the at least one sheave nest is engaged with the at least one receiver. In this exemplary embodiment or another exemplary embodiment, the spreader beam further comprises: an upper securement assembly operatively engaged with the at least one sheave nest at the at least one upper sheave nest aperture and with the at least one receiver at the upper receiver aperture. In this exemplary embodiment or another exemplary embodiment, the at least one receiver further comprises: a cover plate covering an access port where at least a portion of the at least one lower securement assembly is accessible.

[0007] In another aspect, the invention provides a method of repairing a spreader beam, comprising: providing the spreader beam, wherein the spreader beam comprises: a main frame adapted to suspend a load of material; a first receiver operatively engaged with the main frame; a second receiver operatively engaged with the main frame; a first sheave nest removably engaged with the first receiver; a second sheave nest removably engaged with the second receiver; a third sheave nest; separating at least the first sheave nest from the first receiver; and installing a third sheave nest with the first receiver. In this exemplary embodiment or another exemplary embodiment, the third sheave nest may be different from the first sheave nest or the second sheave nest, or may be the first sheave nest or the second sheave nest that has been rebuilt or repaired. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave from the first receiver further comprises: removing at least one securement assembly from at least the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the method of repairing a spreader beam further comprises: removing a cover plate from the first receiver; revealing an access port defined in the first receiver; and accessing at least a portion of the at least one securement assembly inside a chamber defined in the first receiver through the access port. In this exemplary embodiment or another exemplary embodiment, the step of removing the at least one securement assembly further comprises: disengaging at least one fastener from an end cap of the at least one securement assembly; disengaging the at least one fastener from a pin of the at least one securement assembly; and separating the end cap from the pin. In this exemplary embodiment or another exemplary embodiment, the step of removing the at least one securement assembly further comprises: removing the pin from the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the step of separating the first sheave nest further comprises: removing at least another securement assembly from the at least the first sheave nest and the first receiver, wherein the at least one another securement assembly is spaced apart from the at least one securement assembly. In this exemplary embodiment or another exemplary embodiment, the step of installing the third sheave nest with the first receiver further comprises: resting the third sheave nest with the first receiver; and engaging the third sheave nest to at least the first receiver with at least one securement assembly. In this exemplary embodiment or another exemplary embodiment, the fourth sheave nest may be different from the first sheave nest or the second sheave nest, or it may be the first sheave nest or the second sheave nest that has been rebuilt or repaired. In this exemplary embodiment or another exemplary embodiment, the method of repairing a spreader beam further comprises: separating the second sheave nest from the second receiver; and installing

a fourth sheave nest with the second receiver. In another aspect, an exemplary embodiment of the present disclosure may provide a sheave nest comprising at least plurality of nest plates interspersed with a plurality of sheaves rotatably mounted to the plurality of nest plates. In another aspect, an exemplary embodiment of the present disclosure may provide a sheave nest comprising at least one nest flange; at least one nest plate; a lower sheave nest aperture defined in at least one of the at least one nest flange and the at least one nest plate; and at least one sheave rotatably engaged with at least one of the at least one nest flange and the at least one nest plate. In this exemplary embodiment or another exemplary embodiment, a sheave nest may be rebuilt or repaired by removing and replacing at least one sheave. In this exemplary embodiment or another exemplary embodiment, the step of removing at least one sheave further comprises loosening and removing at least one axle bolt from the axle. In this exemplary embodiment or another exemplary embodiment, the axle is removed from the sheave nest by pulling on the axle cap.

[0008] In yet another aspect, the disclosure provides a method of reconditioning a spreader beam, comprising: disassembling the spreader beam; maintaining an arcuate orientation of a plurality of suspension cables, wherein the plurality of suspension cables is engaged with a first plurality of sheaves of the spreader beam; removing the first plurality of sheaves from the plurality of suspension cables; replacing the first plurality of sheaves with a second plurality of sheaves, wherein the second plurality of sheaves are different from the first plurality of sheaves. In this exemplary embodiment or another exemplary embodiment, the method further comprises: installing the plurality of suspension cables into the second plurality of sheaves; and reassembling the spreader beam with the second plurality of sheaves. In this exemplary embodiment or another exemplary embodiment, maintaining the arcuate orientation of the at least one plurality of suspension cables further comprises: installing at least one clamp to the plurality of suspension cables at a distance away from the first plurality of sheaves. In this exemplary embodiment or another exemplary embodiment, In this exemplary embodiment or another exemplary embodiment, the method further comprises: installing the plurality of suspension cables into the second plurality of sheaves; removing the at least one clamp from the plurality of suspension cables; and reassembling the spreader beam with the second plurality of sheaves. In this exemplary embodiment or another exemplary embodiment, method further comprises: disassembling the first plurality of sheaves of the spreader beam from a sheave nest. In this exemplary embodiment or another exemplary embodiment, the method further comprises: disassembling a sheave nest of a spreader beam from a main frame of the spreader beam; and disassembling the first plurality of sheaves of the spreader beam from the sheave nest. In this exemplary embodiment or another exemplary embodiment, the method further comprises: securing a plurality of restraints to the main frame; and stabilizing the main frame, by the plurality of restraints, as the first and second sheave nests are separated from the main frame. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing at least a first securement assembly from the sheave nest and a receiver of the spreader beam; and removing at least a second securement assembly from the sheave nest and the receiver. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing an axle from the sheave nest and the receiver; and separating the sheave nest from the receiver. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing a cover plate from the receiver; revealing an access port defined in the receiver; and accessing at least a portion of the at least the first securement assembly inside a chamber defined in the receiver through the access port. In this exemplary embodiment or another exemplary embodiment, the method further comprises: disengaging at least one fastener from an end cap of the at least the first securement assembly; disengaging the at least one fastener from a pin of the at least the first securement assembly; and separating the end cap from the pin. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing the pin from a first sheave nest and a first receiver. In this exemplary embodiment or another exemplary embodiment, the method further

comprises: removing at least a second securement assembly from at least the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the further comprises: disengaging at least one fastener from an end cap of the at least the first securement assembly; disengaging the at least one fastener from a pin of the at least the first securement assembly; and separating the end cap from the pin.

[0009] In another aspect, the present disclosure provides a method of reconditioning a spreader beam, comprising: maintaining an arcuate orientation of a first plurality of suspension cables wherein the first plurality of suspension cables is engaged with a first plurality of sheaves of the spreader beam; maintaining an arcuate orientation of a second plurality of suspension cables wherein the second plurality of suspension cables is engaged with a second plurality of sheaves of the spreader beam; disassembling the spreader beam; and removing the first and second pluralities of sheaves from the first and second pluralities of suspension cables. In this exemplary embodiment or another exemplary embodiment, the method further comprises: stabilizing the main frame, by a plurality of restraints, as the first and second sheave nests are separated from the main frame. In this exemplary embodiment or another exemplary embodiment, the method further comprises the step of maintaining the arcuate orientation of a first plurality of suspension cables further comprises: engaging a first clamp to the first plurality of suspension cables, wherein each suspension cable of the first plurality of suspension cables is spaced apart from one another; wherein the step of maintaining the arcuate and orientation of the second plurality of suspension cables further comprises: engaging a second clamp to the second plurality of suspension cables, wherein each suspension cable of the second plurality of suspension cables is spaced apart from one another. In this exemplary embodiment or another exemplary embodiment, the method further comprises: replacing the first plurality of sheaves with a third plurality of sheaves, wherein the third plurality of sheaves are different from the first plurality of sheaves; and replacing the second plurality of sheaves with a fourth plurality of sheaves, wherein the fourth plurality of sheaves are different from the second plurality of sheaves. In this exemplary embodiment or another exemplary embodiment, the method further comprises: replacing a first sheave nest with a third sheave nest wherein the third sheave nest is different from the first sheave nest; and replacing a second sheave nest with a fourth sheave nest wherein the fourth sheave nest is different from the second sheave nest. In this exemplary embodiment or another exemplary embodiment, the method further comprises: securing the third sheave nest to a first receiver using at least a first securement assembly and securing the fourth sheave nest to a second receiver using at least a second securement assembly. In this exemplary embodiment or another exemplary embodiment, the method further comprises: engaging the third sheave nest to at least the first receiver with at least one securement assembly; and securing the third sheave nest to the first receiver with at least a second securement assembly. In this exemplary embodiment or another exemplary embodiment, the method further comprises: maintaining an arcuate orientation of the first group of suspension cables of the plurality of suspension cables; maintaining an arcuate orientation of the second group of suspension cables of the plurality of suspension cables;

[0010] In yet another aspect, the present disclosure provides a method of reconditioning a spreader beam, comprising: engaging at least a first clamp to a first group of suspension cables of a plurality of suspension cables of the spreader beam; installing at least a second clamp to a second group of suspension cables of the plurality of suspension cables of the spreader beam; disassembling the spreader beam; suspending a first plurality of sheaves from a first support with the first group of suspension cables of the plurality of suspension cables; suspending a second plurality of sheaves from a second support with the second group of suspension cables of the plurality of suspension cables; installing a first sheave nest with the first plurality of sheaves; installing a second sheave nest with the second plurality of sheaves; removing at least the first clamp from a first group of suspension cables of the plurality of suspension cables; and removing at least the second clamp from a second group of suspension cables of the plurality of suspension cables.

[0011] In yet another aspect, the disclosure provides a method of repairing a spreader beam, comprising: providing the spreader beam, wherein the spreader beam comprises: a main frame adapted to suspend a load of material; a first receiver operatively engaged with the main frame; a second receiver operatively engaged with the main frame; a first sheave nest removably engaged with the first receiver, wherein the first sheave nest is engaged with a first plurality of sheaves; a second sheave nest removably engaged with the second receiver wherein the second sheave nest is engaged with a second plurality of sheaves; engaging at least a first clamp to a first group of suspension cables of a plurality of suspension cables; installing at least a second clamp to a second group of suspension cables of a plurality of suspension cables; suspending the first and second pluralities of sheaves from a support with a plurality of suspension cables; wherein a first plurality of suspension cables is engaged with the first plurality of sheaves; wherein a second plurality of suspension cables is engaged with second plurality of sheaves; separating the respective sheave nests from the respective receivers wherein the suspension cables remain engaged with the sheaves; separating the respective pluralities of sheaves from the respective sheave nests wherein the suspension cables remain engaged with the sheaves; separating the respective pluralities of sheaves from respective pluralities of suspension cables; removing at least the first clamp from a first group of suspension cables of the plurality of suspension cables; and removing at least the second clamp from a second group of suspension cables of the plurality of suspension cables. In this exemplary embodiment or another exemplary embodiment, the method further comprises, securing the suspension cables wherein the suspension cables remain attached to the crane or upper support. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest from the first receiver further comprises: removing at least a first securement assembly from at least the a first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest from the first receiver further comprises: removing at least a second securement assembly from at least the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the method further comprises removing at least one axle from the at least the first sheave nest. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest from the first receiver further comprises: removing at least a first securement assembly from at least the second sheave nest and the second receiver. In this exemplary embodiment or another exemplary embodiment, the method further comprises removing at least one axle from the at least the second sheave nest. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing a cover plate from the first receiver; revealing an access port defined in the first receiver; and accessing at least a portion of the at least one securement assembly inside a chamber defined in the first receiver through the access port. In this exemplary embodiment or another exemplary embodiment, the step of removing the at least one securement assembly further comprises: disengaging at least one fastener from an end cap of the at least one securement assembly; disengaging the at least one fastener from a pin of the at least one securement assembly; and separating the end cap from the pin. In this exemplary embodiment or another exemplary embodiment, the step of removing the at least one securement assembly further comprises: removing the pin from the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest further comprises: removing at least a second securement assembly from the at least the first sheave nest and the first receiver; wherein the at least a second securement assembly is spaced apart from the at least a first securement assembly. In this exemplary embodiment or another exemplary embodiment, the method further comprises installing a third plurality of sheaves with a first plurality of suspension cables. In this exemplary embodiment or another exemplary embodiment, the step of installing the third sheave nest with the first receiver further comprises: resting the third sheave nest with the first receiver; and engaging the third sheave nest to at least the first receiver with at least one securement assembly. In this exemplary embodiment or another

exemplary embodiment, the method further comprises: separating the second sheave nest from the second receiver; and installing a fourth sheave nest with the second receiver.

[0012] Another aspect of the disclosure provides a method of repairing a spreader beam, comprising: providing the spreader beam, wherein the spreader beam comprises: a main frame adapted to suspend a load of material; a first receiver operatively engaged with the main frame; a second receiver operatively engaged with the main frame; a first sheave nest removably engaged with the first receiver, wherein the first sheave nest is engaged with a first plurality of sheaves; a second sheave nest removably engaged with the second receiver, wherein the second sheave nest is engaged with a second plurality of sheaves; suspending the first and second pluralities of sheaves from a support with a plurality of suspension cables; wherein a first group of suspension cables of the plurality of suspension cables is engaged with the first plurality of sheaves; wherein a second group of suspension cables of the plurality of suspension cables is engaged with the second plurality of sheaves; separating the first sheave nest from the first receiver; wherein the first group of suspension cables remains engaged with the first plurality of sheaves; and separating the second sheave nest from the second receiver, wherein the second plurality of suspension cables remains engaged with the second plurality of sheaves. In this exemplary embodiment or another exemplary embodiment, the method further comprises: separating the first plurality of sheaves from the first sheave nest, wherein the first group of suspension cables remains engaged with the first plurality of sheaves. In this exemplary embodiment or another exemplary embodiment, the method further comprises: separating the first plurality of sheaves from the first group of suspension cables. In this exemplary embodiment or another exemplary embodiment, the method further comprises: securing a plurality of restraints to the main frame; and stabilizing the main frame, by the plurality of restraints, as the first and second sheave nests are separated from the main frame. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest from the first receiver further comprises: removing at least a first securement assembly from the first sheave nest and the first receiver; and removing at least a second securement assembly from at least the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing at least one axle from the at least the first sheave nest. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the second sheave nest from the second receiver further comprises: removing at least a first securement assembly from at least the second sheave nest and the second receiver; and removing at least a second securement assembly from at least the second sheave nest and the second receiver. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing at least one axle from the at least the second sheave nest. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing a cover plate from the first receiver; revealing an access port defined in the first receiver; and accessing at least a portion of the at least one securement assembly inside a chamber defined in the first receiver through the access port. In this exemplary embodiment or another exemplary embodiment, the step of removing the at least one securement assembly further comprises: disengaging at least one fastener from an end cap of the at least one securement assembly; disengaging the at least one fastener from a pin of the at least one securement assembly; and separating the end cap from the pin. In this exemplary embodiment or another exemplary embodiment, the step of removing the at least one securement assembly further comprises: removing the pin from the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the method further comprises: removing at least one axle from the first sheave nest and a first plurality of sheaves. In this exemplary embodiment or another exemplary embodiment, the method further comprises separating the first plurality of sheaves from the first group of suspension cables; and separating the second plurality of sheaves from the second group of suspension cables. In this exemplary embodiment or another exemplary embodiment, the method further comprises: installing a third plurality of sheaves to a first group of a plurality of

suspension cables to provide suspended sheaves. In this exemplary embodiment or another exemplary embodiment, the method further comprises: installing the third plurality of sheaves in at least one sheave nest to form a plurality of nested sheaves. In this exemplary embodiment or another exemplary embodiment, the method further comprises: installing at least one axle through the plurality of nested sheaves and the at least one sheave nest wherein the plurality of nested sheaves are suspended in a plurality of suspension cables to provide a suspended sheave nest. In this exemplary embodiment or another exemplary embodiment, the method further comprises: installing a suspended sheave nest in a receiver of a spreader beam. In this exemplary embodiment or another exemplary embodiment, the method further comprises securing the sheave nest to the receiver using at least one securement assembly. In this exemplary embodiment or another exemplary embodiment, the method further comprises: securing the sheave nest to the receiver using at least a second securement assembly. In this exemplary embodiment or another exemplary embodiment, the step of installing the third sheave nest with the first receiver further comprises: resting the third sheave nest with the first receiver; and engaging the third sheave nest to at least the first receiver with at least one securement assembly.

[0013] Still another aspect of the disclosure provides a method of repairing a spreader beam, comprising: providing the spreader beam, wherein the spreader beam comprises: a main frame adapted to suspend a load of material; a first receiver operatively engaged with the main frame; a second receiver operatively engaged with the main frame; a first sheave nest removably engaged with the first receiver, wherein the first sheave nest is engaged with a first plurality of sheaves; a second sheave nest removably engaged with the second receiver wherein the second sheave nest is engaged with a second plurality of sheaves; suspending the first and second pluralities of sheaves from a support with a plurality of suspension cables; wherein a first group of suspension cables of the plurality of suspension cables is engaged with the first plurality of sheaves; wherein a second group of suspension cables of the plurality of suspension cables is engaged with the second plurality of sheaves; separating the first sheave nest from the first receiver; wherein the first group of suspension cables remains engaged with the first plurality of sheaves; and separating the second sheave nest from the second receiver, wherein the second plurality of suspension cables remains engaged with the second plurality of sheaves.

[0014] In still another aspect, the disclosure provides a method of repairing a spreader beam, comprising: providing the spreader beam, wherein the spreader beam comprises: a main frame adapted to suspend a load of material; a first receiver operatively engaged with the main frame; a second receiver operatively engaged with the main frame; a first sheave nest removably engaged with the first receiver, wherein the first sheave nest is engaged with a first plurality of sheaves; a second sheave nest removably engaged with the second receiver wherein the second sheave nest is engaged with a second plurality of sheaves; suspending the first and second pluralities of sheaves from a support with a plurality of suspension cables; wherein a first group of suspension cables of the plurality of suspension cables is engaged with the first plurality of sheaves; wherein a second group of suspension cables of the plurality of suspension cables is engaged with the second plurality of sheaves; separating the first sheave nest from the first receiver; separating the second sheave nest from the second receiver, wherein the second plurality of suspension cables remains engaged with the second plurality of sheaves; wherein the first group of suspension cables remains engaged with the first plurality of sheaves; and separating the second sheave nest from the second receiver, wherein the second plurality of suspension cables remains engaged with the second plurality of sheaves.

[0015] Yet another aspect of the disclosure is a method of repairing a spreader beam, comprising: providing the spreader beam, wherein the spreader beam comprises: a main frame adapted to suspend a load of material; a first receiver operatively engaged with the main frame; a second receiver operatively engaged with the main frame; a first sheave nest removably engaged with the first receiver, wherein the first sheave nest is engaged with a first plurality of sheaves; a second

sheave nest removably engaged with the second receiver wherein the second sheave nest is engaged with a second plurality of sheaves; suspending the first and second pluralities of sheaves from a support with a plurality of suspension cables; wherein a first plurality of suspension cables is engaged with the first plurality of sheaves; wherein a second plurality of suspension cables is engaged with second plurality of sheaves; separating the respective pluralities of sheaves from the respective sheave nests wherein the suspension cables remain engaged with the sheaves; separating the respective sheave nests from the respective receivers wherein the suspension cables remain engaged with the sheaves; and separating the respective pluralities of sheaves from the suspension cables. In this exemplary embodiment or another exemplary embodiment, the method further comprises securing the suspension cables wherein the suspension cables remain attached to the crane or upper support. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest from the first receiver further comprises: removing at least a first securement assembly from at least the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest from the first receiver further comprises: removing at least a second securement assembly from at least the first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the method further comprises: engaging at least a first clamp to a first group of suspension cables of the plurality of suspension cables and installing at least a second clamp to a second group of suspension cables of the plurality of suspension cables.

[0016] In still another aspect, the present disclosure provides a method of repairing a spreader beam, comprising: providing the spreader beam, wherein the spreader beam comprises: a main frame adapted to suspend a load of material; a first receiver operatively engaged with the main frame; a second receiver operatively engaged with the main frame; a first sheave nest removably engaged with the first receiver, wherein the first sheave nest is engaged with a first plurality of sheaves; a second sheave nest removably engaged with the second receiver wherein the second sheave nest is engaged with a second plurality of sheaves; engaging at least a first clamp to a first group of suspension cables of a plurality of suspension cables; installing at least a second clamp to a second group of suspension cables of a plurality of suspension cables; suspending the first and second pluralities of sheaves from a support with a plurality of suspension cables; wherein a first plurality of suspension cables is engaged with the first plurality of sheaves; wherein a second plurality of suspension cables is engaged with second plurality of sheaves; separating the respective pluralities of sheaves from the respective sheave nests wherein the suspension cables remain engaged with the sheaves; separating the respective pluralities of sheaves from the suspension cables; and separating the respective sheave nests from the respective receivers; and removing at least the first clamp from a first group of suspension cables of the plurality of suspension cables; and removing at least the second clamp from a second group of suspension cables of the plurality of suspension cables.

[0017] In this exemplary embodiment or another exemplary embodiment, the method further comprises securing the suspension cables wherein the suspension cables remain attached to the crane or upper support. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest from the first receiver further comprises: removing at least a first securement assembly from at least the a first sheave nest and the first receiver. In this exemplary embodiment or another exemplary embodiment, the step of separating at least the first sheave nest from the first receiver further comprises: removing at least a second securement assembly from at least the first sheave nest and the first receiver.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] Sample embodiments of the present disclosure are set forth in the following description, are shown in the drawings and are particularly and distinctly pointed out and set forth in the appended claims.

[0019] FIG. 1 (FIG. 1) is a perspective view of a spreader beam in accordance with one aspect of the present disclosure, wherein the spreader beam is carrying a load.

[0020] FIG. 2 (FIG. 2) is a top, front, right side isometric perspective view of the spreader beam as shown in FIG. 1.

[0021] FIG. 3 (FIG. 3) is a top, front, right side isometric perspective view of a main frame of the spreader beam.

[0022] FIG. 4 (FIG. 4) is a top plan view of the main frame of the spreader beam.

[0023] FIG. 4A (FIG. 4A) is an enlargement of the highlighted region of FIG. 4.

[0024] FIG. 5 (FIG. 5) is a front elevation view of the main frame of the spreader beam.

[0025] FIG. 6 (FIG. 6) is a cross-sectional view taken along line 6-6 in FIG. 5.

[0026] FIG. 7 (FIG. 7) is a cross-sectional view taken along line 7-7 in FIG. 5.

[0027] FIG. 8 (FIG. 8) is a top, front, right side perspective view of a sheave nest of the spreader beam shown in FIG. 2.

[0028] FIG. 9 (FIG. 9) is a front elevation view of the sheave nest.

[0029] FIG. 10 (FIG. 10) is a top plan view of the sheave nest.

[0030] FIG. 11 (FIG. 11) is an exploded view of a lower securement assembly of the spreader beam.

[0031] FIG. 12 (FIG. 12) is an exploded view of an upper securement assembly of the spreader beam.

[0032] FIG. 13A (FIG. 13A) is an operational view of the spreader beam, wherein a cover plate of the spreader beam is removed from the main frame.

[0033] FIG. 13B (FIG. 13B) is another operational view similar to FIG. 13A, but fasteners of the lower securement assemblies are loosened and removed from pins and the end caps of the lower securement assemblies.

[0034] FIG. 13C (FIG. 13C) is another operational view similar to FIG. 13B, but the pins and the end caps of the lower securement assemblies are disengaged from one another and removed from the main frame and respective sheave nest.

[0035] FIG. 13D (FIG. 13D) is another operational view of a sheave nest and another portion of a spreader beam, wherein fasteners of upper securement assemblies are being loosened and removed from pins and the end caps of the upper securement assemblies.

[0036] FIG. 13E (FIG. 13E) is another operational view similar to FIG. 13D, but the pins and the end caps of the upper securement assemblies are disengaged from one another and removed from the main frame and respective sheave nest.

[0037] FIG. 13F (FIG. 13F) is another operational view continuing from FIG. 13E, but the respective sheave nest is removed from a receiver of the main frame of the spreader beam.

[0038] FIG. 13G (FIG. 13G) is a bottom, front, right side perspective view of a sheave nest of the spreader beam shown in FIG. 8.

[0039] FIG. 14A (FIG. 14A) is a perspective view of a spreader beam in accordance with one aspect of the present disclosure, wherein the spreader beam is ready to be disassembled.

[0040] FIG. 14B (FIG. 14B) is a perspective view of a spreader beam in accordance with one aspect of the present disclosure, with an expanded view of portions of the spreader beam most involved in rebuilding.

[0041] FIG. 15A (FIG. 15A) is an operational view wherein a cover plate of the spreader beam is removed from the main frame and but fasteners of the lower securement assemblies are loosened and removed from pins and the end caps of the lower securement assemblies.

[0042] FIG. 15B (FIG. 15B) is an operational view similar to FIG. 15A, but the pins and the end caps of the lower securement assemblies are disengaged from one another and removed from the main frame and respective sheave nest.

[0043] FIG. 15C (FIG. 15C) is an operational view of a sheave nest and another portion of the spreader beam, wherein fasteners of upper securement assemblies are being loosened and removed from pins and the end caps of the upper securement assemblies.

[0044] FIG. 15D (FIG. 15D) is an operational view similar to FIG. 15C, where the pins and the end caps of the upper securement assemblies are disengaged from one another and removed from the main frame and respective sheave nest.

[0045] FIG. 16A (FIG. 16A) is a top, front, right side perspective view of a spreader beam with sheave nests removed and suspended by suspension cables.

[0046] FIG. 16B (FIG. 16B) is an expanded view of a sheave nest being removed from a receiver while suspended by suspension cables.

[0047] FIG. 17A (FIG. 17A) is an expanded view of a sheave nest after removal from a receiver while suspended by suspension cables and preparing to lift the sheave nest out of the sheaves.

[0048] FIG. 17B (FIG. 17B) is an operational view similar to that in FIG. 17A, with the sheave nest being pulled out and removed from the sheaves.

[0049] FIG. 18A (FIG. 18A) is a side elevation view of sheaves removed from a sheave nest and suspended in suspension cables.

[0050] FIG. 18B (FIG. 18B) is a side elevation view of sheaves being removed from suspension cables.

[0051] FIG. 18C (FIG. 18C) is a perspective view of the suspension cables remaining substantially in operating configuration after sheaves are removed, wherein clamps are engaged with the suspension cables to maintain the suspension cables at the operating orientation.

[0052] FIG. 19 (FIG. 19) is a method flowchart of reconditioning a spreader beam.

[0053] Similar numbers refer to similar parts throughout the drawings.

DETAILED DESCRIPTION

[0054] Referring to FIG. 1, spreader beam **20** is generally borne or lifted by crane **10** through suspension cables **11**. Spreader beam **20** is used to carry a load such as a ladle **14** through one or more hooks **13**.

[0055] For reference, a spatial axis system including directions **15**, **17**, and **19** correspond generally to the x-, y-, and z-axes, respectively. That is, **15** is generally horizontal, **17** is generally vertical, and **19** is generally the axis pointing toward the reader, or “out of the page.” The x-axis may also be termed “longitudinal,” while the y-axis may be termed “vertical” or “upright.” Such terms are relative and not limiting.

[0056] As best seen in FIGS. 1-3, spreader beam **20** generally includes a left side **20a** that opposes right side **20b**, a front side **20c** that opposes a back side **20d**, and a top side **20e** that opposes a bottom side **20f**. In addition, the spreader beam **20** includes a longitudinal axis in direction **15** that extends between the left side **20a** and the right side **20b**, a transverse axis in direction **19** that extends between the front side **20c** and back side **20d**, and a vertical axis in direction **17** that extends between the top side **20e** and bottom side **20f**. It should be understood that the terms “front”, “rear”, “top”, “bottom”, “right” and “left” are used to describe the orientation of the spreader beam **20** illustrated in the attached figures and should in no way be considered to limit the orientation in which spreader beam **20** may be utilized.

[0057] Referring now to FIGS. 2-4A, the spreader beam **20** includes a main frame **22** and at least one sheave nest **24** that operatively engages with the main frame **22** at a respective end; such engagement between the main frame **22** and the at least one sheave nest **24** is discussed in greater detail below. As best seen in FIG. 3, a belly band **28** (or other reinforcement) may also operatively engage with the main frame **22** to provide axial support along a lengthwise or longitudinal axis of main frame **22** that is parallel with the direction **15**. It should be understood that main frame **22** is

that portion of the spreader beam **20** not including sheave nests **24**, belly band **28** or receivers **34**. [0058] Referring to FIGS. **1** and **5**, the spreader beam further includes one or more engagement points for carrying a load. For example, such an engagement point may be an arch or hook receiver **37** where at least one hook **13** or other load bearing member is removably and/or rotatably engaged. In one embodiment at least one hook **13** is removably and/or rotatably fastened to the spreader beam **20** through a hook pin **25** which holds at least one hook **13** in place. Bushings **36** may also be included to facilitate rotation of at least one hook **13** around hook pin **25**.

[0059] Referring to FIG. **3**, spreader beam **20** includes a pair of receivers **34** where each end of the main frame **22** includes a receiver of the pair of receivers **34** that receives and engages with a respective sheave nest **24**. In the present disclosure, a first receiver **34** of the pair of receivers **34** operatively engages with the main frame **22** at a first end **22a** of the main frame **22**, and a second receiver of the pair of receivers **34** operatively engages with the main frame **22** at a second end **22b** of the main frame **22** opposite to the first end **22a** (see FIG. **3**). For brevity, the following description of a receiver **34** is describing the second receiver **34**; however, it should be understood that such description of the second receiver **34** applies to the first receiver **34**.

[0060] Referring to FIGS. **4A-5**, receiver **34** is partially defined by walls **34a-34g** which includes a first side wall **34a**, a second side wall **34b** opposite first side wall **34a**, a bottom wall **34c**, a top wall **34d** opposite to the bottom wall **34c**, and first and second distal walls **34e** and **34f** that are opposite to a proximal wall **34g**. Side walls **34a** and **34b** are generally vertical, in the xy plane (direction **15** and **17**). In this exemplary embodiment or another exemplary embodiment, each side wall **34a** and **34b** bears a reinforcing box **40a** and **40b** respectively. Bottom wall **34c** is generally horizontal (in the xz plane-directions **15** and **19**) when viewed from the front elevation view (see FIG. **5**). Top wall **34d** is generally also horizontal but may be angled with respect to the xz plane when viewed from the front elevation view (see FIG. **5**). First and second distal walls **34e** and **34f** may be angled relative to one another. Alternatively, one or both of first and second distal walls **34e** and **34f** may be generally vertical, in the yz plane. In this exemplary embodiment or another exemplary embodiment, there is a single distal wall which conceptually includes first and second distal walls **34e** and **34f** adjoined in a single plane. All walls **34a-34g** together with cradle plates **26a-26d** and **31** define a chamber **81** to be discussed herein below.

[0061] In this exemplary embodiment or another exemplary embodiment, referring to FIGS. **4A** and **10**, cradle plates **26a-26d** are arcuate and/or curvilinear when viewed from a front elevation view as in FIG. **5**. Such cradle plates **26a-26d** are engaged with receiver plates **48-1-48-4** by mounting brackets **45a-45d**. As an example, a first cradle plate **26a** is engaged with a first receiver plate **48-1** by a first mounting bracket **45a**. Cradle plates **26a-26d** engage with, and receive sheaves **52a** of sheave nest **24**, when sheave nest **24** is engaged with receiver **34**. In the present disclosure, a first compartment **43a** is defined between first and second cradle plates **26a** and **26b** (see FIGS. **4A** and **6**). Similarly, a second compartment **43b** is also defined between third and fourth cradle plates **26c** and **26d**. In the present disclosure, the first and second compartments **43a**, **43b** are spaced apart from and opposite to one another relative to an axis that is parallel with axis **19**. Between cradle plates **26b** and **26c** is a cradle plate **31** which defines a third compartment **33**. Sheaves **52a** having diameter **52x** engage with cradle plates **26a-26d**. In an embodiment illustrated by FIG. **10**, it is seen that two central sheaves **52b** have a different diameter, represented by diameter **52y**. This diameter difference among sheaves **52a** and **52b** is not required; sheaves **52a** and **52b** can be identical.

[0062] The central cradle plate **31**, having diameter **31x**, accommodates two central sheaves **52b**. In this exemplary embodiment or another exemplary embodiment, sheaves **52** have a diameter **52x** (in inches) of 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, or 50 or values therebetween. In this exemplary embodiment or another exemplary embodiment, sheaves **52b** generally have a diameter **52y** that is greater than diameter **52x** of sheaves **52a**. While not shown in the drawings, it will be evident to the skilled artisan that actual sheave diameter measurements may refer to the outer sheave diameter or the sheave pitch diameter.

[0063] As seen in FIGS. 4A and 5, the diameter 26x of cradle plates 26 is in one embodiment the same or similar to the diameter 52x of a sheave 52a. In this exemplary embodiment or another exemplary embodiment, diameter 26x of cradle plates 26 is greater than diameter 52x of sheaves 52a. Cradle plates 26 accommodate sheaves 52a. In this exemplary embodiment or another exemplary embodiment, cradle plates 26 have a diameter 26x (in inches) of 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, or 50 or values therebetween.

[0064] In this exemplary embodiment or another exemplary embodiment, as best seen in FIG. 10, sheave nest 24 has a width 24x defined as the distance from the outer edge/face of flange 50a to the outer edge of flange 50d. This width 24x in this exemplary embodiment or another exemplary embodiment, may be, (in inches), 48, 50, 52, 54, 55, 56, 58, 60, 62, 64, 65, 66, 70, 72, 74, 75, 76, 78, 80 or other values there between. Axle 54 may have a length 54x (not shown) similar to, equal to or greater than the width 24x, with similar numerical values as noted in the preceding sentence.

[0065] In this exemplary embodiment or another exemplary embodiment, referring to FIG. 4A, receiver plates 48-1 to 48-4 are parallel to side walls 34a and 34b. Receiver plate 48-1 is proximate wall 34a and reinforcing box 40a, while receiver plate 48-2 opposes receiver plate 48-1 and is proximate side wall 34b and reinforcing box 40b. Receiver plate 48-3 is in the interior of receiver 34, and proximate receiver plate 48-1, and opposes interior receiver plate 48-4, which is proximate receiver plate 48-2. Each receiver plate 48-1 through 48-4 has two sides, for example, 48-1a, which faces side wall 34a and 48-1b, which faces side wall 34b.

[0066] In the present disclosure one or both of the first side wall 34a and the second side wall 34b includes one or more apertures which align with respective apertures of sheave nest 24 once the spreader beam 20 is assembled. In particular, first side wall 34a and/or second side wall 34b may define at least one of an upper receiver aperture 30 and a lower receiver aperture 32. As best seen in FIG. 3, upper receiver aperture 30 is also defined in an upper portion of side wall 34a near the top side 20e of spreader beam 20. Owing to symmetry, the same is true for side wall 34b with respect to upper receiver aperture 30. Still referring to FIG. 3, lower receiver aperture 32 is defined in a lower portion of side wall 34a near the bottom side 20f of spreader beam 20 and in reinforcing box 40a. Owing to symmetry, the same is true for side wall 34b and reinforcing box 40b. Owing to symmetry, the same is true for side wall 34b with respect to lower receiver aperture 32.

[0067] As best seen in FIG. 7, each receiver plate 48-1 through 48-4 defines an upper receiver aperture 30 in an upper portion of the receiver plates 48-1 through 48-4 near the top wall 34d of the receiver 34. As best seen in FIG. 7, each receiver plate 48-1 through 48-4 defines a lower receiver aperture 32 in a lower portion of receiver plates 48 near the bottom side 20f of spreader beam 20.

[0068] Referring to FIG. 6, the lower receiver aperture 32 of each receiver plate 48-1 through 48-4 is coaxial with the lower receiver apertures 32 of the first side wall 34a and the second side wall 34b. For the avoidance of doubt, lower receiver apertures 32 are defined in first side wall 34a, second side wall 34b, receiver plates 48-1 through 48-4, and reinforcing boxes 40a and 40b. For the avoidance of doubt, upper receiver apertures 30 are defined in first side wall 34a, second side wall 34b and all receiver plates 48-1 through 48-4. Further referring to FIG. 7, cross members, 42a, 42b, 44a and 44b are each operatively engaged with at least two receiver plates 48 to provide support in direction 15.

[0069] As seen in FIG. 13, a first sheave nest 24 is removably engaged with receiver 34 by at least one securement assembly 60, 70. As best seen in FIG. 13A, a first or lower securement assembly 60 is inserted into and received by the receiver 34 inside the lower receiver apertures 32, and also by a lower sheave nest aperture 49 of sheave nest 24. First receiver aperture 32 has a diameter 32a (see FIG. 6), and lower sheave nest aperture 49 has a diameter 49a (see FIG. 9). In at least one embodiment, it will be appreciated that 32a and 49a are the same diameter. In at least one embodiment, diameters 32a and 49a correspond to the diameter 61a of a pin 61 of the lower securement assembly 60.

[0070] In another embodiment, a second or upper securement assembly 70 may be simultaneously

inserted through upper sheave nest aperture **47** of sheave nest **24** and upper receiver apertures **30** of receiver **34** such that the upper securement assembly **70** removably engages the sheave nest **24** with receiver **34**. In this exemplary embodiment or another exemplary embodiment, when sheave nest **24** is engaged with receiver **34**, upper sheave nest aperture **47** and upper receiver aperture **30** are coaxial with one another. It will be appreciated that in at least one embodiment, lower receiver aperture **32** and lower sheave nest aperture **49** have the same diameter (**32a** and **49a**, respectively), which correspond to the diameter **61a** of a pin **61** of the lower securement assembly **60**.

[0071] Referring to FIG. **11**, lower securement assembly **60** includes an elongate pin **61** having a first end **61a** and a second end **61b** connected by a cylindrical wall **61c**. Pin **61** also defines a diameter **61d** along the entire length of pin **61** between the first end **61a** and the second end **61b**. First end **61a** in one embodiment also includes end plate **61e** which may have diameter **61d** or another diameter, for example greater than diameter **61d**, (not shown). End cap **64** is releasably engaged with pin **61** by at least one fastener **66**. Diameter **61d** of pin **61** in one embodiment may correspond to an inner diameter (not shown) of end cap **64**. The outer diameter (not shown) of end cap **64** may be greater than diameter **61d**.

[0072] The diameter **61d** of pin **61**, lower sheave nest aperture **49** and lower receiver aperture **32** are such that the pin **61** is received by and engages with the sheave nest **24** and receiver **34** through lower sheave nest aperture **49** and lower receiver aperture **32**. In one exemplary embodiment, diameter **61d** of pin **61** is the same as diameter **32a** of lower receiver aperture **32** and diameter **49a** of lower sheave nest aperture **49**. In one embodiment, the diameter **32a** of lower receiver aperture **32** and diameter **49a** of lower sheave nest aperture **49** is 4-10", preferably 5-9", more preferably 6-8" and most preferably 7". Eye bolt **62** is operatively engaged with the first end **61a** of the pin **61**. In one exemplary embodiment, the eye bolt **62** is screwed into first end **61a** of the pin **61**.

[0073] As seen in FIG. **12**, upper securement assembly **70** includes an elongate pin **71** having a first end **71a** and a second end **71b** connected by a cylindrical wall **71c**. Pin **71** also defines a diameter **71d** along the entire length of pin **71** between the first end **71a** and the second end **71b**. First end **71a** in one embodiment also includes end plate **71e** which may have diameter **71a** or another diameter not shown. In one embodiment, diameter **71d** of pin **71** is the same as diameter **30a** of upper receiver aperture **30** and diameter **47a** of upper sheave nest aperture **47**. Pin **71** is received by and engages with the sheave nest **24** and receiver **34**. An eyebolt **72** is also operatively engaged with first end **71a** of pin **71**. In one exemplary embodiment, the eyebolt **72** is screwed into first end **71a** of the pin **71**. End cap **74** is releasably engaged with pin **71** by at least one fastener **76**.

Diameter **71d** of pin **71** in one embodiment may correspond to an inner diameter (not shown) of end cap **74**. The outer diameter (not shown) of end cap **74** may be greater than diameter **71d**. The diameter (not shown) of first end **71a** of pin **71** may in one embodiment be greater than diameter **71d** of pin **71**. It is understood that at least one fastener **66**, **76** may be a bolt. In one embodiment, the diameter **30a** of upper receiver aperture **30** and diameter **47a** of upper sheave nest aperture **47** are 3-7", preferably 4-6", and most preferably 5". The securement assemblies **60** and **70** are generally elongate and generally have a cross section that is circular, that is, the securement assemblies **60** and **70** are cylindrical, however other shapes are possible. Lower securement assembly **60** and associated parts may have the same or different dimensions as upper securement assembly **70** and its respective associated parts.

[0074] FIG. **13A** depicts sheave nest **24** engaged with receiver **34** in its operational configuration. That is, sheave nest **24** is securely attached to receiver **34** through lower securement assembly **60** and upper securement assembly **70**. In some embodiments, sheave nest **24** may be attached to receiver **34** by the lower securement assembly **60** or the upper securement assembly **70** if desired.

[0075] As seen in FIGS. **13B-13D**, lower securement assembly **60** is received by and releasably engages with sheave nest **24** at the lower sheave nest aperture **49** and with receiver **34** at the lower receiver apertures **32**. In one exemplary embodiment, lower securement assembly **60** may be received by only an outer nest plate subassembly that includes nest flanges **50a** and **50b** together

with nest plate **46a**, or it may be further received by nest plate **46b**. Alternatively, lower securement assembly **60** may be received by and releasably engage with all nest plates **46a-46d** or any subgrouping thereof and/or all nest flanges **50a-50d** or any subgrouping thereof.

[0076] As seen in FIG. **13E**, upper securement assembly **70** is received by and releasably engages with a sheave nest **24** at upper sheave nest aperture **47** and with receiver **34** at the upper receiver apertures **30**. In one exemplary embodiment, upper securement assembly **70** may be received by only nest plate **46b** or nest plate **46c**. Alternatively, upper securement assembly **70** may be received by and releasably engage with nest plates **46b** and **46c**.

[0077] As seen in FIGS. **8** and **9**, sheave nest **24** comprises main weldment **41**, to which nest flanges **50a-50d** and nest plates **46a-46d** are integrally connected which forms the main structure of sheave nest **24**. In one embodiment, and as best seen in FIG. **10**, a first nest plate **46a** is positioned between first and second nest flanges **50a**, **50b**, while a second nest plate **46d** is positioned between third and fourth nest flanges **50c**, **50d**. Nest plates **46c** and **46d** may be also positioned between other nest flanges **50** and/or similar components (not shown) when required for structural purposes; however, such configuration is not preferred and/or required herein. Other configurations are possible.

[0078] Still referring to sheave nest **24**, a lower sheave nest aperture **49** is defined in each nest flange **50a-50d** and in each nest plate **46a-46d**. Sheave nest **24** may also define a upper sheave nest aperture **47** in at least a second nest plate **46b** and third nest plate **46c**. In one exemplary embodiment, an upper sheave nest aperture **47** may alternatively be defined in at least one nest plate **46a-46d** and in at least one nest flange **50a-50d** or any subset thereof.

[0079] Further referring to FIG. **8** and also referring to FIG. **10**, at least one sheave **52** is interspersed between, and rotatably engaged with, nest plates **46a-46d** and/or nest flanges **50a-50d**. In one embodiment, a plurality of sheaves **52** is present. In this or another embodiment, a first set of sheaves of the plurality of sheaves **52** is between nest plates **46a** and **46b**, a second set of sheaves of the plurality of sheaves **52** is between nest plates **46b** and **46c**, and a third set of sheaves of the plurality of sheaves **52** is between nest plates **46c** and **46d**. In the foregoing discussion, it is presumed that at least one of nest plates **46a-46d** is surrounded by nest flanges **50**. In the present disclosure, a sheave assembly **53** comprises a plurality of sheaves **52** between two consecutive nest plates, for example **46b** and **46c**. In one embodiment, the number of sheaves **52** between nest plates **46a** and **46b** is equal to the number between nest plates **46c** and **46d**. The same or a different number of sheaves **52** may be between nest plates **46b** and **46c**. In various embodiments, a sheave assembly **53** may include two, three, four, five, six, or more sheaves **52**.

[0080] As seen in FIG. **9**, a sheave nest **24** comprises upper and lower ends **24a** and **24b**, proximal end **24c** and distal end **24d**. Such terms are taken relative to the spreader beam **20**. Sheave nest ends **24a-24d** are generally rounded. Upper sheave nest end **24a** may be taken to mean the upper portion of any of nest flanges **50a-50d** or the upper portion of any of nest plates **46a-46d** or both. Proximal end **24c** is generally taken to be the proximal curve of nest plates **46**, however it may include the proximal end of nest flanges **50**. Axle **54** operatively engages with and is housed inside at least one nest flange **50a**, at least one nest plate **46a**, and at least one sheave **52**. In one embodiment, axle **54** extends through all nest flanges **50** and all nest plates **46** that are present in sheave nest **24**. Axle **54** forms an axis of rotation about which sheaves **52** rotate during operation of sheave nest **24** while engaged with spreader beam **20**. Axle **54** comprises axle cap **54a** and is fastened by one or more axle bolts **54b** and has a diameter **54c**. In this exemplary embodiment or another exemplary embodiment, axle diameter **54c** is generally greater than diameter **49a** of lower sheave nest aperture **49**. Similarly, lower receiver aperture diameter **49a** is generally greater than diameter **47a** of upper sheave nest aperture **47**. In one embodiment, one or more end axle cap protectors **55** protects and generally protrudes beyond the profile of axle cap **54a**. In this exemplary embodiment or another exemplary embodiment, axle diameter **54c** is about 6-10", preferably about 7-9" and most preferably about 8".

[0081] In a top-down view, FIG. 10 shows the generally parallel structure of nest plates 46, nest flanges 50, and sheaves 52, which are also further defined by various dimensions. For example, nest plate thicknesses 46e-46h are generally greater than respective nest flange thicknesses 50e-50h, but need not be. The thicknesses of nest plates 46e-46h are generally the same, but need not be. The thicknesses of nest flanges 50e-50h are generally the same, but need not be. Further, nest plate gap 51b (that between nest plate 46b and 46c) is generally greater than nest plate gap 51a (between nest plates 46a and 46b) and nest plate gap 51c (between nest plates 46c and 46d). The width 52c of a sheave 52 is determined by the thickness or diameter (not shown) of the cable 11. The width 52c of sheaves 52 may be the same or different. In this exemplary embodiment or another exemplary embodiment, width 52c may be 2", 2.5", 3", 3.5", 4", 4.5", 5", 5.5", 6", 6.5", 7", 7.5", 8" or other values therebetween.

[0082] The diameter 52a of sheaves 52 may be the same or different. In this exemplary embodiment or another exemplary embodiment, sheave diameter 52a may be (in inches) 24, 25, 26, 27.5, 28, 30, 32, 32.5, 34, 35, 36, 37.5, 38, 40, 42, 42.5, 44, 45, 46, 47.5, 48, or 50 or other values therebetween. There may be (but need not be) a gap 53a, between individual sheaves 52 or between sheave assemblies 53.

[0083] A sheave nest 24 may further comprise at least one lifting lug 27 by which to lift the sheave nest 24 out of a spreader beam 20. Sheave nest 24 may also further comprise one or more equalizer lifting lugs 57 where sheave nest 24 is removably engaged with an equalizer 56. A an equalizer lifting lug 57 may comprise at least one bushing 58.

[0084] FIG. 13A depicts sheave nest 24 engaged with receiver 34 in its operational configuration. That is, sheave nest 24 is securely attached to receiver 34 through securement assembly 60 and securement assembly 70.

[0085] In this exemplary embodiment or another exemplary embodiment, a single securement assembly 60 is used to releasably engage sheave nest 24 with receiver 34 through lower sheave nest aperture 49 and lower receiver aperture 32.

[0086] FIG. 13A also depicts the operation where sheave nest 24 is about to be removed from receiver 34. Cover plate 84, which is removably engaged with distal wall 34f, is pulled in direction 77 (such action is denoted by an arrow labeled 77) after removing at least one fastener 78, for example, a bolt, from cover plate 84. FIG. 13B then shows the end view of receiver 34 after cover plate 84 is removed to reveal access port 80. In this exemplary embodiment or another exemplary embodiment, access port 80 provides access to the lower securement assemblies 60, including pins 61, end caps 64, and fasteners 66, which are located in chamber 81. Access port 80 has a height 80a and a width 80b, which may be the same or different. Height 80a and width 80b may independently be about 18", 20", 24", 30", 35", 36", 40", 42", 45", 48", or other values therebetween.

[0087] To continue the removal of a sheave nest 24 from receiver 34, fasteners 66 are moved in directions 67a and 67b to disengage them from pins 61 and end caps 64. In this operation, fasteners 66 are simply loosened and/or unscrewed from pins 61 by a tool or device that may apply torque on said fasteners 66. In FIG. 13C, it is seen that end caps 64a and 64b are removed from pins 61 by pulling in directions 65a and 65b respectively. Pins 61 may be pulled by grasping eye bolts 62. Pins 61 are pulled in directions 63a and 63b, respectively. At this point, sheave nest 24 is separable from the receiver 34 at the lower sheave nest aperture 49 and at the lower receiver aperture 32.

[0088] Once the lower securement assemblies 60 are removed from the sheave nest 24 and the receiver 34, such operations of removing the upper securement assemblies 70 from the sheave nest 24 and receiver 34 are now undertaken. FIG. 13D shows the removal of fasteners 76 from pins 71 and end caps 74 by unscrewing the fasteners 76 from the pins 71 and end caps 74 and pulling in directions 73a or 73b as appropriate. FIG. 13E then shows end caps 74a and 74b being pulled away from pins 71 by pulling pins 71 in directions 79a and 79b, respectively. Pins 71 may be pulled by grasping eyebolts 72. At this point, sheave nest 24 is separable from the receiver 34 at the upper sheave nest aperture 47 and at upper receiver aperture 30.

[0089] Upon such removal of the lower securement assemblies **60** and the upper securement assemblies **70**, sheave nest **24** is disengaged from receiver **34** and may be removed from the receiver **34**. FIG. **13F** shows chain **12** attached to lifting lug **27** pulling sheave nest **24** out of receiver **34**. Such pulling action is denoted by the arrow labelled **82**. Chain **16** may be attached to spreader beam **20** at lifting lug **27** to hold spreader beam **20** while sheave nest **24** is pulled out of receiver **34**.

[0090] Referring to FIGS. **9** and **13G**, following removal of sheave nest **24** out of receiver **34**, sheave nest **24** may have at least one shipping bolt **35a**, inserted through holes **39** and through at least one hole **59** in or through at least one sheave **52**, or at least one hole **39** below at least one sheave **52** to prevent separation of such at least one sheave **52** from sheave nest **24** and/or to prevent rotation of such at least one sheave **52** during shipping, for example, to a repair or rebuilding facility. The at least one shipping bolt **35a** is secured with at least one shipping nut **35b**.

[0091] At this point, a new, rebuilt, or repaired sheave nest **24** is then inserted into receiver **34**. In this exemplary embodiment or another exemplary embodiment, reassembly is effected by reversing the steps in the preceding paragraphs relating to, and depicted in, FIGS. **13A** through **13G**.

[0092] A sheave nest **24** may be rebuilt or repaired by removing and replacing at least one sheave **52**. Referring to FIGS. **9** and **13A**, to remove at least one sheave **52**, axle **54** is loosened and removed by loosening and removing axle bolts **54b** from axle **54**. Then, axle **54** is removed from sheave nest **24** by pulling on axle cap **54a**. This operation is similar to that depicted in FIGS. **13B** and **13C**. Axle cap protectors **55** may be required to be removed prior to the removal of axle **54**.

[0093] FIG. **14A**, together with the figures that follow, depict a method of reconditioning, that is repairing or rebuilding, a spreader beam **20**. In this view, and throughout the method depicted by FIGS. **14A** through **18C**, it is noted that suspension cables **11** remain affixed to the crane **10** or other upper support, as also shown in FIG. **1**. That is, cables **11** remain in, or are held in, substantially the same configuration during the disassembly and reassembly of spreader beam **20** as during operation of spreader beam **20**. This method of removing sheave nests **24** and sheaves **52** is accomplished without the need to fully unwind the suspension cables **11**, thereby saving time and cost to rebuild or repair sheaves **52** and/or sheave nests **24**.

[0094] Spreader beam **20** may be placed on the ground or other stable surface (as denoted by a dashed-dot-dot line in FIG. **14A**) in preparation for removal of the sheave nests **24** from main frame **22**. Prior to placing spreader beam **20** on the ground, any load or cargo that spreader beam **20** was carrying (for example ladle **14** from FIG. **1**), is removed. Alternatively, spreader beam **20** can be stabilized or supported by one or more securement members such as chains **16a** and/or chains **16b**, or any manner other than that by which suspension cables **11** are engaged with sheaves **52** (see FIG. **14B**). For example, chains **16a** may be engaged with lifting lug **38**, or chains **16b** may be engaged with belly band loops **28a** (FIG. **14B**). Such support is in anticipation of pulling sheave nests **24** out of spreader beam **20**. At this point, suspension cables **11** remain engaged with sheaves **52** while sheaves **52** remain engaged with sheave nests **24**.

[0095] FIG. **14B** depicts the operation where sheave nest **24** is to be removed from receiver **34**. While only one end of spreader beam **20** is shown in FIG. **14B**, it will be appreciated that the same operations will be undertaken on the other end of spreader beam **20**. Cover plate **84**, which is removably engaged with distal wall **34f** (shown in FIGS. **13A** and **13B**), is pulled in direction **77** (such action is denoted by an arrow labeled **77**) after removing at least one fastener **78**, for example, a bolt, from cover plate **84**.

[0096] FIG. **15A** shows the end view of receiver **34** after cover plate **84** is removed to reveal access port **80**. In this exemplary embodiment or another exemplary embodiment, access port **80** provides access to the lower securement assemblies **60**, including pins **61**, end caps **64**, and fasteners **66**, which are located in chamber **81**. Lower securement assemblies **60** and upper securement assemblies **70** are loosened and removed from sheave nest **24** and receiver **34**.

[0097] Referring to FIG. **15A**, to continue the removal of sheave nest **24** from receiver **34**,

fasteners **66** are moved in directions **67a** and **67b** to disengage them from pins **61** and end caps **64**. In this operation, fasteners **66** are simply loosened and/or unscrewed from pins **61** by a tool or device that may apply torque on said fasteners **66**. In FIG. **15B**, it is seen that end caps **64a** and **64b** are removed from pins **61** by pulling in directions **65a** and **65b** respectively. Pins **61** may be pulled by grasping eye bolts **62**. Pins **61** are pulled in directions **63a** and **63b**, respectively. At this point, sheave nest **24** is separable from the receiver **34** at the lower sheave nest aperture **49** and at the lower receiver aperture **32**.

[0098] Once the lower securement assemblies **60** are removed from the sheave nest **24** and the receiver **34**, the operations of removing the upper securement assemblies **70** from the sheave nest **24** and receiver **34** are next undertaken. FIG. **15C** shows the removal of fasteners **76** from pins **71** and end caps **74** by unscrewing the fasteners **76** from the pins **71** and end caps **74** and pulling in directions **73a** or **73b** as appropriate. FIG. **15D** shows end caps **74a** and **74b** being pulled away from pins **71** by pulling pins **71** in directions **79a** and **79b**, respectively. Pins **71** may be pulled by grasping eyebolts **72**. At this point, sheave nest **24** is separable from the receiver **34** at the upper sheave nest aperture **47** and at upper receiver aperture **30**.

[0099] Upon removal of the lower securement assemblies **60** and the upper securement assemblies **70**, sheave nest **24** is disengaged from receiver **34** and may be removed from the receiver **34**. FIGS. **16A** and **16B** show that suspension cables **11** remain engaged with sheaves **52** when pulling sheave nests **24** out of receivers **34**. Such pulling action is denoted by the arrow labelled **82** in FIG. **16B**. The foregoing represents an exemplary embodiment. Another exemplary embodiment (not shown) envisions pulling sheaves **52** out of sheave nests **24** prior to pulling sheave nests **24** out of receivers **34**.

[0100] Referring to FIGS. **17A** and **17B**, following removal of sheave nest **24** from receiver **34**, sheave nest **24** may be removed from (pulled out of) sheaves **52**. Such pulling motion is indicated by the arrow labelled **85**. In such step, sheave nest **24** may be supported with a securement member such as chain **16** that is engaged with lifting lug **57** of sheave nest **24**. After sheave nest **24** is removed from receiver **34**, while still suspended by chain **16**, the sheave nest **24** is removed from the sheaves **52** (see FIG. **17B**). To remove sheave nest **24** from the sheaves **52** while the sheaves **52** are still suspended in suspension cables **11**, axle **54** is loosened and removed by loosening and removing axle bolts **54b** from axle **54**. Then, axle **54** is removed from sheave nest **24** by pulling on axle cap **54a**. This operation is similar to that depicted in FIGS. **13B** and **13C**. Axle cap protectors **55** may be removed prior to the removal of axle **54**.

[0101] FIG. **18A** shows sheaves **52** suspended on suspension cables **11** after sheave nest **24** is separated from sheaves **52**. FIG. **18B** depicts the sheaves **52** being removed from suspension cables **11**. Sheaves **52** may be pulled out of suspension cables **11** by a chain **16**, a cable, or other similar components that may secure to sheaves **52** for removing such sheaves **52** from suspension cables **11**. Such pulling motion is indicated by the arrow labelled **86**. FIG. **18C** shows suspension cables **11** with all elements under discussion (spreader beam **20**, sheave nests **24** and sheaves **52**) removed.

[0102] FIG. **18C** also shows optional clamps **18** attached to or engaged with suspension cables **11**. Clamps **18** are designed to surround and secure a plurality of cables in a desired configuration. Clamps **18** may hold suspension cables **11** together, anywhere along a substantially straight and vertical portion of suspension cables **11**, or in one embodiment as close to the bottom of such cables as will allow the removal and reinstallation of sheave nests **24** and sheaves **52** to suspension cables **11**. In this way, suspension cables **11** are maintained in substantially the same configuration as during operation of spreader beam **20**. It will be appreciated that, in one embodiment, one spreader beam **20** includes two ends, that is, two sheave nests **24** and two pluralities of sheaves **52**, as well as two pluralities of suspension cables **11**, (see FIGS. **14A** and **16A**). Other configurations are possible. FIGS. **16B** through **18B** depict one such sheave nest **24** and one plurality of sheaves **52** held by one plurality of suspension cables **11**. FIG. **18C** depicts one plurality of suspension

cables **11** after all portions of spreader beam **20** have been removed.

[0103] Through use of clamps **18**, suspension cables **11** are maintained in substantially the same configuration as when engaged with sheaves **52** or in the fully operational configuration as shown in FIG. **1** or FIG. **14A**. That is, as best shown in FIG. **18C**, suspension cables **11** remain “parallel” to one another, while maintaining an arcuate configuration at their bottom, which is the portion previously engaged with sheaves **53**. In this context, “parallel” means that the suspension cables remain spaced apart and substantially in the same configuration as during operation of the spreader beam **20**. While only two suspension cables **11** are shown in FIGS. **18A** and **18B**, it will be appreciated that any number of suspension cables **11** may be present. In FIG. **18C**, and as in earlier figures, ten suspension cables **11** are shown in a substantially “parallel” configuration as previously engaged with a corresponding sheave nest **24**.

[0104] To reassemble or recondition spreader beam **20**, new or rebuilt sheaves **52** and/or new or rebuilt sheave nests **24** are installed to spreader beam **20**. In one example, such old or original sheaves **52** that were installed with sheave nests **24** may be referred to herein as first and second pluralities of sheaves while the new, rebuilt, or repaired sheaves **52** that reinstalled with sheave nests **24** may be referred to herein as third and fourth pluralities of sheaves **52** to differentiate such sheaves **52**. Generally, this may be accomplished by reversing the steps depicted in FIGS. **14A** through **18C**, with the exception that clamps **18** are removed as the final step of reassembly.

[0105] During this entire reassembly process, suspension cables **11** remain in substantially their operating orientation, that is, substantially parallel and having an arcuate configuration. Once sheaves are installed to suspension cables **11**, suspension cables **11** remain engaged with sheaves **52**. Finally, clamps **18** are removed from suspension cables **11**.

[0106] In further detail, to begin the rebuilding of spreader beam **20**, new, rebuilt or reconditioned sheaves **52** are positioned to be suspended in suspension cables **11** (FIG. **18A**). At this step, the clamps **18** are engaged with the suspension cables **11** to maintain the spacing and/or positioning of the suspension cables **11** that match with the spacing of the sheaves **52**. It should be noted that if clamps **18** are not used, operators may simply place and/or align each suspension cable **11** with the respective sheave **52** prior to a respective sheave nest **24** being engaged therewith. Next, sheaves **52** receive sheave nests **24** (FIG. **17B**). Then, axle **54** and axle cap **54a** are installed to sheave nest **24** and sheaves **52**. Axle bolts **54b** are inserted and tightened. Sheave nests **24** are then inserted into receivers **34**. Lower securement assembly **60** and upper securement assembly **70** then are engaged with sheave nest **24** and receivers **34**.

[0107] In particular, lower sheave nest apertures **49** of the at least one sheave nest **24** are positioned to be coaxial with the lower receiver apertures **32** of the at least one receiver **34** in order to engage the at least one sheave nest **24** with the at least one receiver **34**. Further, upper receiver apertures **30** are positioned to be coaxial with the upper sheave nest apertures **47** in order to engage the at least one sheave nest **24** with the at least one receiver **34**. The actions in the preceding two sentences are shown in FIGS. **17A** and **16B**.

[0108] Lastly, irrespective of the order in which sheaves **52** and sheave nests **24** are reassembled to one another, or to suspension cables **11**, clamps **18** are removed from the plurality of suspension cables **11** to result in a rebuilt and/or reconditioned spreader beam ready for operation.

[0109] The method above may be accomplished without unwinding the suspension cables **11** from the spreader beam **20** or refeeding the suspension cables **11** into the spreader beam **20** as they may remain suspended from above, engaged with a crane, or otherwise supported. This advantage greatly reduces the time and cost to rebuild a spreader beam.

[0110] It should be understood herein that chains **16** may simply be any restraint or similar elements of the like for restraining any component of the spreader beam **20**, including main frame **22**, at least one sheave nest **24**, and at least one plurality or set of sheaves **52**. As such, chains **16** and similar elements of the like disclosed herein may be referred to herein as restraints or restraining members.

[0111] FIG. 19 provides a flowchart representing a disclosed method 90 of reconditioning a spreader beam, which substantially discloses the following steps:

A method of reconditioning a spreader beam comprising: [0112] 90a—starting with an assembled spreader beam; [0113] 90b—wherein a plurality of suspension cables is engaged with a first plurality of sheaves of the spreader beam; [0114] 90c—disassembling the spreader beam; [0115] 90d—maintaining an operating orientation of a plurality of suspension cables; [0116] 90e—removing the first plurality of sheaves from the plurality of suspension cables; and [0117] 90f—and replacing the first plurality of sheaves with a second plurality of sheaves, wherein the second plurality of sheaves are different from the first plurality of sheaves.

[0118] Various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different from illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

[0119] While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

[0120] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0121] The articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims (if at all), should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or

list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0122] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0123] While components of the present disclosure are described herein in relation to each other, it is possible for one of the components disclosed herein to include inventive subject matter, if claimed alone or used alone. In keeping with the above example, if the disclosed embodiments teach the features of A and B, then there may be inventive subject matter in the combination of A and B, A alone, or B alone, unless otherwise stated herein.

[0124] As used herein in the specification and in the claims, the term “effecting” or a phrase or claim element beginning with the term “effecting” should be understood to mean to cause something to happen or to bring something about. For example, effecting an event to occur may be caused by actions of a first party even though a second party actually performed the event or had the event occur to the second party. Stated otherwise, effecting refers to one party giving another party the tools, objects, or resources to cause an event to occur. Thus, in this example a claim element of “effecting an event to occur” would mean that a first party is giving a second party the tools or resources needed for the second party to perform the event, however the affirmative single action is the responsibility of the first party to provide the tools or resources to cause said event to occur.

[0125] When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly coupled” to another feature or element, there are no intervening features or elements present. Although described or shown with respect to one embodiment, the features and elements so described or shown can apply to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

[0126] Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “above”,

“behind”, “in front of”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”, “horizontal”, “lateral”, “transverse”, “longitudinal”, and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

[0127] Although the terms “first” and “second” may be used herein to describe various features/elements, these features/elements should not be limited by these terms, unless the context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed herein could be termed a second feature/element, and similarly, a second feature/element discussed herein could be termed a first feature/element without departing from the teachings of the present invention.

[0128] An embodiment is an implementation or example of the present disclosure. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, are not necessarily all referring to the same embodiments.

[0129] If this specification states a component, feature, structure, or characteristic “may”, “might”, or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

[0130] As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For example, a numeric value may have a value that is $\pm 0.1\%$ of the stated value (or range of values), $\pm 1\%$ of the stated value (or range of values), $\pm 2\%$ of the stated value (or range of values), $\pm 5\%$ of the stated value (or range of values), $\pm 10\%$ of the stated value (or range of values), etc. Any numerical range recited herein is intended to include all sub-ranges subsumed therein.

[0131] Additionally, the method of performing the present disclosure may occur in a sequence different from those described herein. Accordingly, no sequence of the method should be read as a limitation unless explicitly stated. It is recognizable that performing some of the steps of the method in a different order could achieve a similar result.

[0132] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively.

[0133] To the extent that the present disclosure has utilized the term “invention” in various titles or sections of this specification, this term was included as required by the formatting requirements of

word document submissions pursuant the guidelines/requirements of the United States Patent and Trademark Office and shall not, in any manner, be considered a disavowal of any subject matter. [0134] In the foregoing description, certain terms have been used for brevity, clearness, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed.

[0135] Moreover, the description and illustration of various embodiments of the disclosure are examples and the disclosure is not limited to the exact details shown or described.

Claims

1. A spreader beam comprising: a main frame adapted to suspend a load of material; and at least one sheave nest removably engaged with the main frame.
2. The spreader beam of claim 1, further comprising: at least one receiver operatively engaged with the main frame and configured to receive the at least one sheave nest.
3. The spreader beam of claim 2, wherein the at least one receiver comprises: at least one side wall operatively engaged with the main frame; and at least one cradle plate operatively engaged with the at least one side wall; wherein the at least one cradle plate is configured to receive the at least one sheave nest.
4. The spreader beam of claim 3, wherein the at least one receiver further comprises: a lower receiver aperture defined in the at least one side wall.
5. The spreader beam of claim 4, wherein the at least one sheave nest comprises: at least one nest flange; at least one nest plate; a lower sheave nest aperture defined in at least one of the at least one nest flange and the at least one nest plate; and at least one sheave rotatably engaged with at least one of the at least one nest flange and the at least one nest plate.
6. The spreader beam of claim 5, wherein the lower sheave nest aperture of the at least one sheave nest is coaxial with said lower receiver aperture of the at least one receiver when the at least one sheave nest is engaged with the at least one receiver.
7. The spreader beam of claim 6, further comprising: at least one lower securement assembly operatively engageable with the at least one sheave nest at the at least one lower sheave nest aperture and with the at least one receiver at the lower receiver aperture.
8. The spreader beam of claim 7, wherein the at least one sheave nest further comprises: at least one another nest plate operatively engaged with the at least one nest flange; and at least one upper sheave nest aperture defined by at least one of the at least one nest flange and the at least one another nest plate.
9. The spreader beam of claim 8, wherein the at least one receiver further comprises: at least another side wall operatively engaged with the main frame and opposite to the at least one side wall; and an upper receiver aperture defined in at least one of the at least one side wall and the at least another side wall.
10. The spreader beam of claim 9, wherein the upper receiver aperture is coaxial with the at least one upper sheave nest aperture when the at least one sheave nest is engaged with the at least one receiver.
11. The spreader beam of claim 10, further comprising: an upper securement assembly operatively engaged with the at least one sheave nest at the at least one upper sheave nest aperture and with the at least one receiver at the upper receiver aperture.
12. The spreader beam of claim 11, wherein the at least one receiver further comprises: a cover plate covering an access port where at least a portion of the at least one lower securement assembly is accessible.
13. A method of repairing a spreader beam, comprising: providing the spreader beam, wherein the spreader beam comprises: a main frame adapted to suspend a load of material; a first receiver

operatively engaged with the main frame; a second receiver operatively engaged with the main frame; a first sheave nest removably engaged with the first receiver; a second sheave nest removably engaged with the second receiver; a third sheave nest; separating at least the first sheave nest from the first receiver; and installing the third sheave nest with the first receiver.

14. The method of claim 13, wherein the step of separating at least the first sheave nest from the first receiver further comprises: removing at least one securement assembly from at least the first sheave nest and the first receiver.

15. The method of claim 14, further comprising: removing a cover plate from the first receiver; revealing an access port defined in the first receiver; and accessing at least a portion of the at least one securement assembly inside a chamber defined in the first receiver through the access port.

16. The method of claim 14, wherein the step of removing the at least one securement assembly further comprises: disengaging at least one fastener from an end cap of the at least one securement assembly; disengaging the at least one fastener from a pin of the at least one securement assembly; and separating the end cap from the pin.

17. The method of claim 16, wherein the step of removing the at least one securement assembly further comprises: removing the pin from the first sheave nest and the first receiver.

18. The method of claim 14, wherein the step of separating at least the first sheave nest further comprises: removing at least another securement assembly from the at least the first sheave nest and the first receiver; wherein the at least one another securement assembly is spaced apart from the at least one securement assembly.

19. The method of claim 13, wherein the step of installing the third sheave nest with the first receiver further comprises: resting the third sheave nest with the first receiver; and engaging the third sheave nest to at least the first receiver with at least one securement assembly.

20. The method of claim 19, further comprising: separating the second sheave nest from the second receiver; and installing a fourth sheave nest with the second receiver.
